

Is physics discovering reality, or constructing models of it?

Physics, or all science, has an objective to formulate a few comprehensive principles to explain certain phenomena in real life (Weidner and Brown, 2019). Guided by this objective, with the collaborative work of physicists, physics has discovered reality, translating natural events into comprehensible concepts to humans. However, for theories to represent reality, simplified scientific methods and reasoning built upon assumptions must construct a mathematical or experimental model to justify concepts (Rogers, 2012). So the question arises: is physics discovering reality or constructing models of it? But, this question only arises due to a misconception. Discovering realities and constructing models of reality are not two discrete concepts; rather, models are catalysts for discovering reality.

The book of Physics was originally put together by Aristotle around 350 BCE in Ancient Greece. Although he was not the first to philosophize about physics, he was the first to derive the concept of physics as a distinct field of knowledge. For Aristotle, physics was important as it described reality, “understanding natural things by identifying their causes.” His intention shows that the birthright of physics was to explain natural revelations, ultimately discovering reality in the sense of learning the actions of the real world (Anthony J.P. Kenny and Amadio, 2019).

Through time and the diversification of science, contemporary physics is defined as the science of matter, motion, and energy. But its objective remains similar: “the formulation of a few comprehensive principles that bring together and explain all such disparate phenomena,” (Albert Einstein on space-time | Britannica, 2019). This pursuit of unifying principles inherently requires simplification and abstraction, as complex natural phenomena must be reduced into manageable forms.

In this sense, Einstein’s quote reflects the circumstantial importance of modeling and how it brings forth the blueprint of ideas, leading to scientific breakthroughs. The formulation

of principles involves the necessary use of modelling to simplify reality. Limiting variables are necessary to show a direct relationship between different elements of the system; even the simplest and most fundamental laws assume the absence of other variables to state that one parameter directly affects the other. For example, Newton's first law of motion states that an object at rest remains at rest, and an object in motion remains in motion at constant velocity unless acted on by an unbalanced force (Hall, 2024). This law holds true only within inertial frames of reference, which is an ideal assumption, making the law strictly valid only under idealized conditions. According to Einstein's general theory of relativity, perfect global inertial frames of reference do not generally exist due to curvature in spacetime (Overduin, 2007). Even in situations that seem to be an inertial frame of reference, such as in empty space, spacetime curvature produces tidal effects, meaning perfect inertial behavior cannot extend over finite regions. And even in an everyday context, the rotation of the Earth and acceleration due to gravity make a perfectly inertial frame near impossible (sciencedemonstrations.fas.harvard.edu, n.d.).

However, these assumptions behind laws do not mean they are not discovering reality. Even though the rule cannot be perfectly applied in real-life situations, it still holds significant meaning. In accordance with the law of inertia, local inertial frames can be assumed over short times and distances as accelerations become negligible. This shows how modelling removes these variables, which shows a clear relationship through the law. Ultimately, models do not completely define reality –the use of them enables the development of theories that are utilized to create meaningful impact in understanding and implementation.

If physics did not utilize models to discover reality, the limited understanding generated would limit advancements. As modelling allows idealizing conditions, isolating variables, and simplifying other parameters, without modelling relationships would only be

qualitatively stated and have limited explanations. For instance, one could observe that a pendulum swings periodically and that increasing the length of the string increases the time taken for oscillation. However, this qualitative analysis would fail to come up with a definitive law due to many variables present in real life, such as air resistance and friction. Through modelling, physicists simplify the situation by idealizing the conditions, like massless strings, negligible air resistance, and importantly, small-angle approximations. These idealizations derive a simple equation $T = 2\pi\sqrt{\frac{L}{g}}$, simply revealing the square root relationship of the length of the pendulum and its period (Lumen Learning, 2023). Without modelling through assumptions, not only would the equation be far more complex, but the relationship would remain qualitative rather than mathematical.

In a larger lens, without modelling, physics would fail to continue to discover reality in the long term. Although qualitative reasoning can describe simple relationships, it does not develop specific technology to further discover more complex and less intuitive concepts. An example of this could be found in electromagnetism. Before James Clerk Maxwell, laws regarding electricity and magnetism existed, but they were largely distinctive. Forces between charges could be seen through attraction and repulsion, leading to Coulomb's law (courses.lumenlearning.com, n.d.). And with magnetism, the induction of a magnetic field due to current and the other way around, shown in Faraday's law of induction, could also be qualitatively defined with magnetic forces attracting or repelling (Britannica, 2013). However, these qualitative expressions hindered the unification of two ideas into one. So, a key step was introducing modelling to the problem, an electric and magnetic field. Fields in space cannot be directly observed, as only the forces between two charges can be observed. And from this model, Maxwell derived four key mathematical equations that revealed the fact that changes in each field induced one another. This revelation discovered that light is an electromagnetic wave, which the discovery would have otherwise been impossible without

modelling in the form of the idea of fields and the creation of mathematical expressions (www.phys.unsw.edu.au, n.d.).

This case of the unification of electromagnetism and the discovery of light as an electromagnetic wave exemplifies how qualitative reasoning will only discover parts of reality and that modelling becomes necessary to continue discovering reality. The application of the electric and magnetic fields also demonstrates this, such as through the use of mass spectrometers and wireless communication, which enables further discovery of reality with more accurate measurements and applications of theories (Broad Institute, 2010), (Fundamentals of Wireless Communication, n.d.).

However, there are apparent limitations to the claim that physics discovers reality through modelling, as some areas of physics suggest that physics does not discover reality but only produces useful models of observations. An example of this is the wave-particle duality in modern physics (Davis, 2012). Particles such as electrons behave both as waves and particles depending on the conditions (Lumen Learning, 2024). Of course, the duality is incompatible in classical physics, as waves are continuous while particles are discrete. This ambiguity seemingly contradicts the idea of physics discovering reality, as it can only provide different models for different situations rather than fully revealing what electrons are.

Although this limitation clearly exists, it overlooks the role of a model as a tool to discover reality. Despite the incompatibility of the two models, both wave and particle models consistently produce accurate and testable predictions, indicating that they are not arbitrary but describing real physical behavior. So, rather than viewing these two models incompatible, it can be understood that they provide different perspectives of the same event, each capturing partial truths depending on the conditions of the observation. The usage of multiple models to describe single phenomena does not obstruct the discovery of reality but rather helps understanding through simplifying the complex occurrences into understandable

frameworks. So, this rather reinforces the idea that physics discovers reality through the construction of coherent models.

In conclusion, the distinction between physics discovering reality and constructing models of it is ultimately misleading. Throughout the history of physics, starting from Aristotle to modern physics, models played a crucial role in translating physical phenomena into understandable information. While it is true that these models do not perfectly replicate reality by relying on simplifying assumptions, they describe the relationship between variables, which would otherwise be unclear due to the complexity of the natural world. Therefore, physics uses models as a tool to discover reality.

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